Motors and generators

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish electrical-to-mechanical transfer from mechanical-to-electrical transfer.

SUCCESS CRITERIA - I can...

1. Trace energy through a motor and a generator.
2. Identify magnetic interactions in rotation or induction.
3. Include less useful energy transfers.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A consistently drawn magnetic-field map has 3 times as many field lines through the same small perpendicular area at A as at B. What does the denser spacing indicate?

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\eta = P_{useful}/P_{input}$; $E_{input} = E_{useful} + E_{losses}$. Motor: electrical to mechanical; generator: mechanical to electrical. Count losses and any stored-energy change.

Worked reasoning: An idealized motor receives 20 J electrically and delivers 12 J mechanically; 8 J goes to other transfers. A generator runs the energy conversion in the opposite useful direction.

Your turn: Draw separate motor and generator energy-flow diagrams.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and interpret the model

Complete two energy chains: battery → motor → turning axle, and turning axle → generator → lamp. Label each input and useful output.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

04 Apply the physics

A model-car motor receives 100 J of electrical energy and delivers 60 J of useful mechanical energy. Complete the flow model, calculate efficiency and identify the energy transferred to other forms.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

05 Explain your reasoning

A motor drives a generator that feeds the motor. Explain why real losses prevent the loop from running forever without external input.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An electric field of 6 N/C points left. Draw the force direction on +4 C and on -4 C, then compare sizes. Draw field and force arrows separately for each charge.

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3

07 Exit check

Which device converts mechanical input into electrical output?

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A motor draws the same electrical power in two designs, but one delivers more mechanical power. Compare efficiencies and name a measurement needed.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3